Cover Letter

To: Editorial Office, *npj Quantum Information* (Nature Portfolio) **From:** Aleaddin Özer (Corresponding Author) · ORCID 0000-0001-9389-5357 **E-mail:** ozer@e2esolutions.tech **Date:** 1 June 2026

Dear Editors,

We would like *npj Quantum Information* to consider our manuscript *"Three Axes of Quantum Risk: A Unified Observability Model for PQC, QKD, and Crypto-Agility"* as an Original Research Article.

The motivation is operator-side. The post-quantum migration is being driven now by NIST FIPS 203/204/205 and by the NSA CNSA 2.0 schedule, but the cryptographic inventory tools operators rely on still classify assets along one dimension: PQ-resistant or not. We argue this is structurally insufficient. An asset's posture also depends on the channel through which its keys arrive (where QKD becomes a measured property, not a separate research topic), and on how quickly its primitive can be replaced before the deadline. The paper formalises these as Axis A, Axis C, and Axis G, combines them with the deadline horizon into a per-asset score $q(asset, t)$, and ships an open reference platform that observes all three.

What we think is interesting for *npj QI* readers

Quantum information research has shaped the underlying primitives — both lattice KEMs and QKD links. The gap our work fills is between the science and the deployment posture: how do you measure what fraction of a real infrastructure benefits from each, and how do you compare two assets when one has a PQ KEM but no QKD link and the other is classical but QKD-protected? Axis C is the part most likely to interest QST/QI readers: a per-session attribution of QKD-derived key material, validated on a working ETSI GS QKD 014 emulator we release alongside the paper, with 13 induced failure modes correctly classified.

The empirical results are modest but reproducible. We measured 43.8% hybrid PQ-KEM adoption (317 of 724 responsive hosts) on a Tranco-top-1k probe. We measured 91% agreement with hand-graded ground truth on an eleven-project crypto-agility pilot drawn from a fifty- project corpus. Every artifact — schema, implementation, emulator, replay corpus, Semgrep ruleset, hand-graded TSV, Tranco snapshot identifier — is open at https://github.com/e2esolutions-tech/sezar under MIT.

A note on APC

Both authors are self-funding this work; we have no external grant or industrial sponsor covering open-access fees. If the journal can offer a discount on the Original Research APC we would gratefully take it, but our intent is to publish in *npj QI* regardless of the outcome of that request.

Submission disclosures

- **Originality.** The manuscript is not under consideration anywhere else and has not been published.
- **Authorship.** Both authors approved the submission. CRediT contributions are stated in §10 of the manuscript.
- **Competing interests.** None declared. E2E Solutions releases Sezar under MIT but earns no direct revenue from this publication.
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- **Data and code.** See the Data and Code Availability block in §10 of the manuscript; everything is on GitHub under MIT.
- **Suggested reviewers.** Five names in suggested-reviewers.md, three of them first authors on cited references. No co-author or advisor relationship with the submitting authors.
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